divisibility tests

A number is divisible by another if, after dividing, there is no remainder.

A number is ...

divisible by	if	Examples
2	the last digit is even	2, 4, 6, ... 122 ... 358 ... 1000
3	the sum of all digits can be divided by 3	261: 2 + 6 + 1 = 9 3672: 3 + 6 +7 + 2 = 18 18: 1 + 8 = 9
4	the last two digits are divisible by 4	1024: 24 ÷ 4 = 6
5	the last digit is 5 or 0	15, 70 ...
6	the last digit is even and the sum of its digits is divisible by 3	7446: 7 + 4 + 4 + 6 = 21
7	there is no divisibility test	
8	the last 3 digits are divisible by 8	75 384: 384 ÷ 8 = 48
9	the sum of its digits is divisible by 9	3123: 3 + 1 + 2 + 3 = 9
10	the number ends in 0	10, 20, 30 ...

Important: No number can be divided by 0.

See factors, remainder

divisible

A number is divisible by another number if, after dividing, there is no remainder.

Example

$72 \div 9 = 8 \quad 72 \div 8 = 9$

Seventy-two is divisible by nine and also by eight.

Nine and eight are factors of seventy-two.

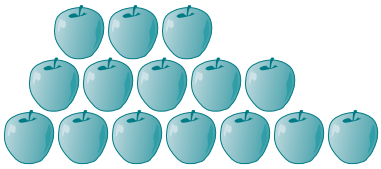
See factors, remainder

division

Division is a mathematical operation which can be interpreted in several different ways:

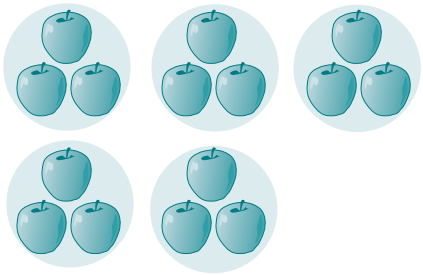
(i) Grouping (quotition).

Example



How many groups of 3 can be made with 15 apples?

The apples are to be placed into groups of equal size, 3 to a group. The problem is to find out how many groups there will be.



$15 \div 3 = 5$

There are 5 groups of 3 apples.

Repeated subtraction is a form of grouping.

division *continued...*